

Aya Squalli Houssaini

SCAN ME



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Experience

HEADSTARTER FELLOWSHIP

SOFTWARE ENGINEERING FELLOW

JULY 2024 - SEPTEMBER 2024

- Built and deployed 5 AI-powered web apps in 5 weeks using React, Next.js, Firebase, Clerk, and Vercel via agile sprints and CI/CD.
- Co-developed a technical support chatbot (Next.js + Gemini API) with a 3-person team.
- Created a SaaS flashcard generator (200+ cards/session) with Stripe paywall and tiered pricing.
- Engineered a RAG-based academic Q&A system using Pinecone, OpenAI GPT-3.5-turbo, and a mock database of 100+ professor reviews.
- Scaled all apps to 50–100+ users, iterating weekly on live feedback for UX and performance.
- Received mentorship from engineers at Google, YC, Stanford, Amazon, and VC-backed startups.
- Finalist in the Rilla Sales Transcripts Hackathon

3DSMART FACTORY

SUMMER INTERN

JULY 2025 – SEPTEMBER 2025

- Researched and prototyped a Federated Learning–based Intrusion Detection System (FL-IDS) for IoT networks.
- Designed a privacy-preserving architecture enabling edge devices to collaboratively train anomaly detection models without sharing raw traffic data.
- Achieved > 95% detection accuracy on real-world dataset (N-BalIoT) while reducing central server load

MCPC PROGRAMMING CONTEST

VOLUNTEER ORGANIZER / VOLUNTEER LEAD

MARCH 2024/ SEPTEMBER 2025

- Assisted in organizing and coordinating the MCPC programming contest, ensuring smooth execution of the event.
- Managed logistics, including participant registration, scheduling, and resource allocation.
- Contributed to the promotion and outreach efforts, increasing participant engagement and event visibility

CAPTURE THE FLAG (CTF) COMPETITIONS

- **Local CTF with 1337**: Competed in and organized a cybersecurity challenge organized by 1337, solving various cryptography, reverse engineering, digital forensics and web security challenges.
- **HackTheBox Meetups** : Participated in a Hack The Box community meetups, engaging in penetration testing challenges and collaborating with security enthusiasts.
- **MCSC CTF at ENSIAS**
- **HackTheBox Meetup with 1337 campuses, 42 Berlin, 42 Wolfsburg and 42 Heilbronn** : Secured 5th place in the CTF competition, solving 6/7 of the problems collaborating in a team of 3.
- **Global HackTheBox Meetup Ambassadors**: 1st prize globally.
- **Kaspersky CTF**: Secured 38th/450 regionally (Middle East, Türkiye and Africa)
- **Nullhat Morocco 2025** : Secured 6th/40 in Morocco, a national competition in collaboration with HackerOne
- **Organizing Local CTF competitions by creating challenges, managing logistics and infrastructure.**
- **ASIS Finals 2025**: 1st in Morocco, 2nd in MENA region and 34th/476 globally
- **UofTCTF 2026**: 24th / 1550 globally playing in the open division
- **IDEH 7th edition / INPT**: 4th place nationally

Skills

- **Programming**: Python, C, C++ , JavaScript, SQL
- **Web Development**: React, Next.js, Nuxt.js, Vue.js (Beginner), HTML5/CSS
- **Cybersecurity & CTF**: Cryptography, Reverse Engineering, Web Exploitation, Forensics, OSINT

Education

College Of Computing-UM6P

STATE ENGINEER IN COMPUTER SCIENCE / CYBERSECURITY MINOR (BAC +5, MASTERS' EQUIVALENT)

EXPECTED GRADUATION: 2029

- **ACM Chapter Member**: Actively participated in the Association for Computing Machinery (ACM) chapter as a membership chair, contributing to events and workshops

Certifications

CS50X: INTRODUCTION TO COMPUTER SCIENCE

Harvard University, edX

COMPLETED: JULY 2024